Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

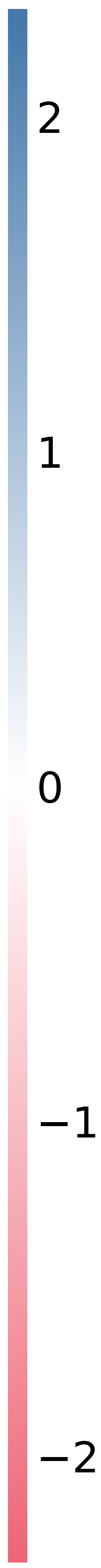
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

KE=1eV=mçĶµåŃĲv^2/2

1eV=1.6ÃĹ10^-19J

æīĢä»¥mçĶµåŃĲv^2=2.1ÃĹ10^-19J

Î»=h/v=hâĪļ(2mçĶµåŃĲKE)

h=6.626ÃĹ10^-34Js, mçĶµåŃĲ=9.11ÃĹ10^-31kg

Î»=6.626ÃĹ10^-34ÃĹâĪļ(2ÃĹ9.11ÃĹ10^-31ÃĹ2.1ÃĹ10^-19)

âīĪ5.27ÃĹ10^-9mâīĪ5.27ÃĹ10^-10cm

</think><answer>

F.5.27ÃĹ10^-9m

</answer>